

Title: Prediction of Halitosis Using Neural Network and Machine Learning Models from Oral and Systemic Health Data

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Short title: NN and ML Prediction of Halitosis

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Abstract

Objective:

To identify key clinical and systemic factors associated with self-reported halitosis using conventional machine learning (ML) and transformer-based neural network (NN) models, focusing on age-specific patterns and explainable artificial intelligence (AI) techniques.

Methods:

A total of 821 patients were included: 134 and 687 with and without self-reported halitosis, respectively. Salivary, oral, systemic, and medication-related variables were analyzed using logistic regression, random forest, XGBoost, and FT-Transformer models. SHapley Additive exPlanations (SHAP), permutation importance, and attention-based interpretation were applied to evaluate feature contributions. Data imbalance was addressed using Synthetic Minority Over-sampling Technique.

Results:

Unstimulated salivary flow rate (UFR) and buffering capacity were the most influential predictors across models. UFR was the most important predictor in both younger (<30 years; SHAP value: +0.213) and older (≥ 50 years; +0.227) adults, whereas buffering capacity exerted a strong negative influence, particularly in young adults (−0.233). Oral mucosal conditions such as stomatitis, burning mouth syndrome, and glossodynia significantly contributed to halitosis prediction in adults aged ≥ 50 years (SHAP value >0.15). Systemic conditions, including hypertension and the cumulative number of systemic diseases, demonstrated modest SHAP values (<0.05), indicating a limited predictive contribution compared to oral health features. The FT-Transformer model showed the highest predictive performance (area under the receiver operating characteristic curve: 0.761 [95% confidence interval: 0.662–0.853]), particularly when oral health features were included. Low and diffuse feature importance was observed in middle-aged individuals (30–49 years old). Explainable AI techniques enabled clinically meaningful interpretation of model predictions, revealing age-specific patterns and complex feature interactions. Oral conditions were consistently stronger predictors of halitosis than systemic or medication-related factors.

Conclusion:

Salivary function and mucosal conditions are key determinants of self-perceived halitosis and their impact varies with age. Transformer-based deep learning models, particularly FT-Transformer, provide superior performance and interpretability for modeling complex oral-systemic interactions. These findings highlight the value of age-stratified explainable AI models for supporting precise oral health assessments and personalized halitosis management.

Keywords: Halitosis, Machine Learning, Transformer Model, Oral Health, Salivary Flow Rate, SHAP Analysis

1. Introduction

Halitosis, or bad breath, is prevalent in individuals of all age groups, with a reported global prevalence of 22–50% depending on the population and methodology of diagnosis [1, 2]. Although often perceived as a minor oral hygiene issue, halitosis can exert profound psychosocial effects, including social withdrawal, diminished self-esteem, and even social isolation [3]. These effects can seriously impair the quality of life and psychological well-being of an individual, making halitosis a clinical and social health issue [4].

The etiology of halitosis is multifactorial, and is typically categorized as intraoral or extraoral. Intraoral causes, such as tongue coating, periodontal disease, caries, and poor oral hygiene, account for most cases [5]. In particular, alterations in salivary function, including reduced flow rate and pH and impaired buffering capacity, promote volatile sulfur compound (VSC) production by anaerobic bacteria, which are a major intraoral cause of halitosis [6, 7]. Salivary function plays a particularly important role in oral homeostasis, and parameters such as optimal unstimulated and stimulated flow rates (UFR and SFR, respectively), pH, and buffering capacity of the saliva are essential for maintaining oral mucosal integrity and preventing oral discomfort [8]. When these functions are impaired, xerostomia can occur, which can exacerbate halitosis. Extraoral causes of halitosis are systemic conditions such as diabetes mellitus (DM), gastrointestinal (GI) disorders, and hepatic or renal disease, but are often under-recognized in clinical practice [9, 10].

Despite the availability of objective tools such as organoleptic scoring and gas chromatography, the diagnosis of halitosis in routine practice often relies on subjective self-assessment or social feedback. These methods are vulnerable to psychological influences such as stress, anxiety, and somatosensory amplification, particularly in cases of pseudo-halitosis or halitophobia [11, 12]. This highlights the need for a more systematic understanding of halitosis and its predictors, particularly in diverse clinical settings. Given this complexity, it is essential to investigate the specific oral, systemic, and behavioral conditions in which individuals may experience halitosis. Understanding the clinical and psychosocial contexts that lead to perceived malodor is not only valuable for patients struggling with distressing symptoms but also critical for clinicians seeking to make informed diagnostic and therapeutic decisions [13]. Identifying these contributing factors may allow for a more accurate differentiation between actual halitosis and its subjective variants, ultimately supporting personalized and effective care.

Machine learning (ML) approaches offer a promising avenue to address this complexity. Conventional ML models such as logistic regression, support vector machine (SVM), and Random Forest, have been used in modeling non-linear relationships and identifying predictors in various oral health domains, including dental caries, periodontal disease, and xerostomia [14–16]. These models are advantageous because of their interpretability and ease of integration into clinical workflows. However, their ability to detect high-dimensional interactions may be limited when dealing with heterogeneous health data combining salivary, oral, systemic, and medication-related variables. By contrast, transformer-based deep learning models have recently emerged as powerful tools for analyzing structured tabular data, demonstrating a

superior performance even in complex feature dependencies [17, 18]. Their attention mechanisms allow the dynamic assessment and interpretation of interactions between features, offering greater flexibility and predictive power [19]. Despite these advantages, the applications of transformer models in halitosis research remain limited. Furthermore, neural network (NN)-based models, although sometimes overlooked in clinical prediction studies, may offer complementary strengths by detecting latent patterns that are not readily accessible to rule-based learners. In this study, we aimed to evaluate and compare several conventional ML and transformer-based models to determine the approaches that can most effectively predict self-reported halitosis across diverse clinical profiles.

In this study, we aimed to identify the clinical and systemic factors most strongly associated with self-reported halitosis by applying several conventional ML and transformer-based NN models. Rather than focusing solely on prediction, our goal was to assess the relative contributions of diverse health variables, including salivary flow rate and buffering capacity, oral mucosal conditions, systemic diseases, and medication use, and examine these associations across age groups. We also evaluated whether ensemble approaches offer improved performance over individual models and assessed the ability of NNs, particularly FT-Transformer, to detect complex feature interactions. To enhance interpretability and clinical relevance, we applied explainable artificial intelligence (AI) techniques, such as SHapley Additive exPlanations (SHAP) analysis, permutation importance, and attention-based interpretation. We hypothesized that oral health-related features, especially those reflecting salivary function and mucosal integrity, would have a stronger predictive value than systemic variables. Moreover, we anticipated age-related variations in predictor importance, reflecting shifts in the pathophysiology and perception of halitosis throughout the lifespan of an individual.

2. Materials and Methods

The research protocol for this study complied with the Declaration of Helsinki and was approved by the Institutional Review Board of Kyung Hee University Dental Hospital in Seoul, South Korea (KHD IRB, IRB No. KH-DT23022). Informed consent was obtained from participants aged ≥ 18 years and from the parents and/or legal guardians of patients aged < 18 years. Two trained dentists evaluated each variable. Inter-rater reliability, as measured by Cohen's kappa, ranged from 0.82–0.85. In cases of disagreement, a consensus was reached through multiple rounds of discussion.

2.1 Participants

In this cross-sectional study, we enrolled 821 patients aged from their teens to 80s and who visited the Department of Oral Medicine and Orofacial Pain at Kyung Hee University Dental Hospital from January 1, 2017, to December 31, 2021 with oral disease and related discomfort

(mean age, 59.36 ± 16.43 years; 234 males and 587 females). Based on the presence of subjective halitosis, participants were categorized into non-halitosis (mean age: 60.38 ± 15.41 years; 192 males and 495 females) and halitosis (mean age: 54.12 ± 20.14 years; 42 males and 92 females) groups. All participants were asked to complete a questionnaire to analyze the presence of subjective halitosis, sex, age, UFR, SFR, presence of systemic diseases, and number of systemic diseases. Patients who did not agree to participate in the study, had incomplete data, or dropped out of the study were excluded.

2.2 Evaluation of Halitosis

Participants were asked to respond to the following dichotomous questions: “Do you feel you have bad breath?” and/or “Have you ever recognized your bad breath based on someone else's comment?” Those who answered ‘yes’ to either question were classified into the halitosis group, indicating self-perceived or socially reported halitosis. Participants who answered “no” to both questions were classified into the non-halitosis group.

2.3 Assessment of Salivary Parameters

All salivary evaluations were performed by skilled dentists who had been trained in a standardized examination protocol. Assessments were performed during the initial visit of the patients. The salivary flow rate was measured using the spitting method and included both UFR and SFR, expressed in mL/min^{1,2}. Additional salivary parameters, including salivary pH and buffering capacity, were evaluated using the GC Saliva Check Buffer kit (GC Corporation). After collecting stimulated saliva, a pipette was used to load the sample on the buffer test strip. The buffering capacity was determined based on the color change of the strip, and was interpreted according to the manufacturer's salivary buffer indicator. Subjective xerostomia was assessed by asking patients to respond dichotomously (yes or no) to the following question, “During the past 4 weeks, have you frequently experienced a dry mouth sensation?”

2.4 Clinical Assessment of Oral Health and Disease Presence

Oral health conditions were assessed and documented by skilled dentists through clinical examinations. The presence of oral diseases was evaluated by two experienced dentists and included eight conditions: stomatitis, recurrent aphthous ulcers (RAU), candidiasis, periodontitis, lichen planus, obstructive sialodochitis, glossodynia, and burning mouth syndrome (BMS). The patients rated their oral discomfort using a visual analog scale (VAS), with scores ranging from 0 to 10. Sticky saliva was defined as frothy or viscous saliva detectable with the naked eye. Oral hygiene status was subjectively classified by the examining clinician into three categories: good, acceptable, or poor. Calculus deposition was subjectively

assessed and recorded as none, moderate, or severe. The presence of oral ulcers was confirmed when visible lesions were detected on the oral mucosa. Oral candidiasis was clinically diagnosed when patients exhibited typical clinical features such as tongue coating, tongue atrophy, or angular cheilitis.

2.5 Evaluation of Systemic Diseases and Medications

The general health status of patients was assessed using a self-reported questionnaire. The presence of systemic diseases and systemic medication use were dichotomously classified as “yes” or “no”. Systemic diseases included hypertension, DM, renal diseases, hyperlipidemia, liver diseases, GI disorders, arthralgia, rheumatism, osteoporosis, cardiovascular diseases (CVD), thyroid disorders (TD), mental disorders, urinary diseases, asthma, and history of cancer surgery. Systemic medications included those prescribed for hypertension, DM, osteoporosis, hyperlipidemia, CVD, TD, GI disorders, urinary diseases, arthritis, and rheumatism; anxiolytics; sleeping pills; and aspirin. For analyses involving the number of systemic diseases or medications, the total count was calculated by assigning a value of “1” to each “yes” response and “0” to each “no” response.

2.6 Categorization of Input Features for Model Training

For training the conventional ML models, all variables were categorized into five domains based on the data described above: (1) Basic information (n = 5): age, sex, UFR, SFR, and subjective xerostomia; (2) Oral health characteristics (n = 6): salivary pH, salivary buffering capacity, VAS score for oral pain, sticky saliva, oral hygiene, and calculus deposition; (3) Oral diseases (n = 8): stomatitis, RAU, candidiasis, periodontitis, lichen planus, obstructive sialodochitis, glossodynia, and BMS, denoted with the prefix “O_”; (4) Systemic diseases (n = 13): hypertension, DM, hyperlipidemia, renal diseases, liver diseases, rheumatism, osteoporosis, CVD, TD, mental disorders, urinary diseases, asthma, and a history of cancer surgery, denoted with the prefix “S_”; and (5) Medications (n = 13): medications for the aforementioned conditions, as well as anxiolytics, sleeping pills, and aspirin, denoted with the prefix “M_”.

2.7 Classifications with Multiple Features using ML Algorithms

2.7.1 Data Description and Splitting

The collected raw data were organized in Excel sheets and subsequently subjected to random splitting before model development. The dataset (n = 821) was organized in Excel and split into training (n = 492), validation (n = 164), and test (n = 165) sets using stratified sampling

(60:20:20) to preserve the class ratios. Class imbalance (Label_1 $\approx$ 16%) was addressed by applying Synthetic Minority Over-sampling Technique (SMOTE) only to the training set (Label_0 = 412, Label_1 = 412), while validation and test sets retained the original distribution. Random seed 42 ensured reproducibility (Figure 1).

2.7.2 Addressing Class Imbalance Through SMOTE

The original dataset exhibited a significant class imbalance, with considerably fewer instances of the minority class (Label_1). To address this issue, SMOTE was applied only to the training data, generating synthetic Label_1 sample by interpolating between the nearest neighbors. This increased the number of minority classes from 80 to 412, resulting in a balanced training set (Label_0 = 412, Label_1 = 412). This strategy improved the ability of the model to learn minority class patterns without overfitting (Figure 1).

2.7.3 Comparative Overview of ML and NN Approaches

In this study, we used tabular data on halitosis to develop and compare two groups of models: four conventional ML models and four NN models: TabTransformer, FT-Transformer, SAINT, and TabNet. Hyperparameter tuning was performed for all models to optimize predictive performance and manage model complexity. Among ML models, XGBoost and Random Forest utilize boosting and bagging strategies, respectively, to aggregate multiple decision trees. CatBoost is particularly effective for handling categorical variables, and SVM can learn complex decision boundaries using kernel functions. Among the NN models, TabTransformer, FT-Transformer, and SAINT use self-attention mechanisms to model interactions among input features. By contrast, TabNet uses stepwise attentive feature selection, providing both interpretability and strong performance in structured data (Figure 2).

2.7.4 Optimization and Benchmarking of ML and NN Models

2.7.4.1 Conventional ML models

We developed and fine-tuned four conventional ML models to classify halitosis using both oral and systemic health data. To address class imbalance, all models were trained on datasets augmented using the SMOTE. Hyperparameter optimization was conducted via cross-validation to maximize the area under the receiver operating characteristic (ROC) curve (AUC). For XGBoost, a 5-fold cross-validation strategy with early stopping (early_stopping_rounds = 10) was employed, yielding the following optimal parameters: n_estimators = 32, max_depth = 7, and learning_rate = 0.2. CatBoost, known for its efficiency in handling categorical variables, was tuned using GridSearchCV with 5-fold cross-validation, resulting in iterations = 200, depth = 7, and learningrate = 0.1. The SVM model underwent hyperparameter tuning via GridSearchCV with 5-fold cross-validation, with C = 10, kernel = 'rbf', and probability = True to facilitate AUC-based evaluation. For Random Forest, GridSearchCV with 5-fold cross-validation was utilized, resulting in a final model with n_estimators = 300, max_depth = None,

and min_samples_split = 2. These optimized conventional models served as baseline classifiers and provided benchmarks for evaluating the performances of transformer-based NN models in subsequent analyses.

2.7.4.2 Transformer-based and NN models

We implemented four NN architectures specifically designed for tabular data. These models leverage attention mechanisms or feature-selection strategies to detect complex feature interactions within structured datasets. All models were trained on SMOTE-augmented data to address class imbalance, and their predictive performance was assessed using the AUC. TabTransformer utilizes embedded categorical and numerical features processed through transformer encoder blocks to model interfeature dependencies. FT-Transformer adopts an architecture similar to stacked encoders, followed by a classification head, offering a streamlined implementation. SAINT applies self-attention across both features and instances. However, in this study, we adopted a feature-wise attention variant to enhance interpretability. TabNet employs sequential feature selection using sparse attention masks, providing both high predictive accuracy and model explainability. All models were implemented with standard hyperparameters, as reported in the original publications, and evaluated against baseline ML models for comparative performance analysis [20-23]. All experimental procedures and codes used in this study are available at the following online repository: [<https://github.com/seongwoojang1123/Neural-Network-and-Machine-Learning-Prediction-of-Halitosis-from-Oral-and-Systemic-Health-Data.git>].

2.8 Statistical Analysis

Descriptive statistics were reported as mean $\pm$ standard deviation for continuous variables and frequency with percentage for categorical variables. Prior to group comparisons, the assumptions of normality and homogeneity of variance were evaluated using the Shapiro–Wilk and Levene tests, respectively. Continuous variables were compared between groups using the independent t-test with Bonferroni correction, whereas categorical variables were analyzed using the chi-square test with Bonferroni correction. Fisher’s exact test was used when the expected frequency in any cell was less than 5. For subgroup analysis, participants were stratified into three age groups based on widely accepted World Health Organization and the U.S. Centers for Disease Control and Prevention classifications: Group 1 (0–29 years), Group 2 (30–49 years), and Group 3 (≥ 50 years) [24, 25]. These categories reflect physiologically significant stages relevant to salivary function and an increased risk of halitosis, thereby supporting epidemiological comparability. Model performance was assessed using ten-fold cross-validation. ROC curves were generated from the aggregated predicted probabilities across the 10 test sets, and the average AUC was calculated with corresponding 95% confidence intervals (CIs). Differences in AUC between groups (age and sex subgroups) were

assessed using DeLong's method. In addition, cutoff values for xerostomia classification were derived from ROC curve analysis. The inter-rater reliability of the risk-of-bias assessment was evaluated using Cohen's kappa coefficient. All statistical analyses were performed using IBM SPSS Statistics (version 26.0; IBM Corp., Armonk, NY, USA), R version 4.3.2 (R Foundation for Statistical Computing, Vienna, Austria), and Python (version 3.11), with relevant libraries including SciPy 1.12 and scikit-learn 1.4. Statistical significance was defined as a two-tailed p-value <0.05, with adjusted thresholds applied for multiple comparisons when appropriate.

3. Results

3.1 Demographics

The mean age of patients in the halitosis group was significantly lower than that in the non-halitosis group (60.38 ± 15.41 vs. 62.97 ± 15.04 years, $p = 0.0001$). When stratified into three age groups (Group 1: <30 years; Group 2: 30–59 years; and Group 3: ≥ 60 years), both the halitosis and non-halitosis groups showed a significantly lower proportion of individuals in Group 1 than in Groups 2 and 3 ($p = 0.0011$). Within the halitosis group, the distribution of participants across age groups was as follows: Group 3 (47.0%), Group 2 (41.1%), and Group 1 (11.9%) ($p = 0.0011$). The proportions of super-elderly individuals (aged ≥ 75 years) were 18.7% in the halitosis group and 20.4% in the non-halitosis group. The male-to-female ratio was 1:2.58 in the halitosis group and 1:2.19 in the non-halitosis group, with no significant difference between the two groups ($p = 0.4890$).

3.2 Salivary Flow Rate and Oral Health Characteristics

Although the mean UFR was lower in the halitosis group than in the non-halitosis group (0.35 ± 0.26 vs. 0.39 ± 0.24 mL/min), the difference was not statistically significant ($p = 0.1182$). In contrast, the SFR was significantly lower in the halitosis group than in the non-halitosis group (1.17 ± 0.56 vs. 1.32 ± 0.72 mL/min, $p = 0.0321$). The presence of subjective xerostomia was higher in the halitosis group than in the non-halitosis group, but the difference was not significant (29.1% vs. 23.3%, $p = 0.1846$). Among the oral health characteristics, there were no significant differences in salivary pH (6.72 ± 0.68 vs. 6.73 ± 0.63), salivary buffering capacity (10.58 ± 2.23 vs. 10.36 ± 2.37), and VAS score for oral pain (5.62 ± 2.97 vs. 5.46 ± 3.32) between the non-halitosis and halitosis groups (all $p > 0.05$). The prevalence of sticky saliva did not differ between the non-halitosis and halitosis groups (29.1% vs. 27.6%, $p = 0.8054$). Approximately 90% of patients in both groups had acceptable to good oral hygiene. Calculus deposition was also low in severe cases (7.2%), with incidence of 7.4% and 6.0% in non-halitosis and halitosis groups, respectively. The proportion of cases in which no calculus deposition was observed in non-halitosis and halitosis groups was 19.7% and 23.9%, respectively, with no significant difference between the groups ($p = 0.4879$) (Table 1).

3.3 Distribution of Oral Diseases

The prevalence of oral diseases was compared between patients with and without halitosis. Among all conditions, oral stomatitis was significantly more common in the halitosis group (26.1%) than in the non-halitosis group (18.0%) ($p = 0.0411$). Although not statistically significant, RAU was observed in 6.0% of patients in the non-halitosis group and 1.5% in the halitosis group ($p = 0.0555$). Oral candidiasis showed a higher prevalence in the non-halitosis group (15.4%) than in the halitosis group (9.7%) ($p = 0.1121$), although the difference was not statistically significant. Similarly, the prevalences of periodontitis (3.2% vs. 3.0%, $p = 1.0000$), oral lichen planus (5.5% vs. 1.5%, $p = 0.0772$), and sialodochitis (0.9% vs. 0.0%, $p = 0.5951$) did not show significant differences between groups. In contrast, prevalences of glossodynia (2.8% vs. 0.0%, $p = 0.1023$) and BMS (34.1% vs. 30.6%, $p = 0.4983$) were comparable between groups without statistical significance (Table 1).

3.4 Distribution of Systemic Diseases and Medications

Contrary to expectations, the total number of systemic diseases was significantly higher in the non-halitosis group than in the halitosis group (1.02 ± 1.11 vs. 0.75 ± 1.06 ; $p = 0.0112$). Hypertension was the only systemic disease that showed a significantly higher prevalence in the non-halitosis group than in the halitosis group (87.8% vs. 12.2%, $p = 0.0228$). The prevalence of DM, hyperlipidemia, liver disease, osteoporosis, CVD, and TD, did not differ significantly between groups (all $p > 0.05$). Hypertension (26.1%), osteoporosis (17.8%), CVD (17.5%), a history of cancer surgery (21.2%), and liver disease (17.2%) were the five most common systemic diseases in the halitosis group. Regarding medication use, no statistically significant differences were found between the halitosis and non-halitosis groups for medications for hypertension, DM, osteoporosis, hyperlipidemia, CVD, TD, GI disorders, urinary tract issues, arthritis, rheumatism, and psychiatric conditions (anxiolytics and sleeping pills) ($p > 0.05$ for all comparisons). Hyperlipidemia medications (25.5%), CVD (24.1%), osteoporosis medications (22.0%), sleeping pills (20.0%), and anxiolytics (19.2%) were the five most frequently used medications in the halitosis group (Supplementary Table 1).

3.5 Comparison of ROC Performance Between Feature Conditions Across ML Models

Mean ROC curves with 95% CI shading were generated for the four conventional ML classifiers. For each model, the AUC and its 95% CI were calculated to assess model performance under two different feature conditions: Condition 1 (C1), which included basic information and oral health characteristics, and Condition 2 (C2), which included basic information and oral diseases. To statistically evaluate performance differences between the two feature conditions, DeLong's test was applied based on five independent 5-fold CV repetitions. The resulting p -values are annotated above each ROC plot, as shown in Figure 3A.

Among the ML models, CatBoost and Random Forest yielded the highest AUCs under C2 (0.729 and 0.731, respectively), significantly outperforming their C1 counterparts (CatBoost: AUC = 0.614, $p = 0.025$; Random Forest: AUC = 0.644, $p = 0.018$). XGBoost also showed significantly improved performance under C2 (AUC = 0.705) compared with that under C1 (AUC = 0.628; $p = 0.014$). In contrast, SVM did not show a statistically significant difference between the two conditions (AUC = 0.555 vs. 0.680; $p = 0.117$), although a numerical improvement in the AUC was observed. Figure 3B presents the overlay of all 10-fold mean ROC curves for both feature conditions across the four models, further illustrating the consistent improvement in predictive performance when incorporating oral disease features (C2). These results suggest that the inclusion of oral disease variables meaningfully enhanced model performance for halitosis classification, particularly when used in tree-based ensemble models such as XGBoost, CatBoost, and Random Forest.

3.6 Performance of Conventional ML Models in Halitosis Prediction

Table 2 presents the AUC performances of the four ML models applied to seven sequentially expanded variable sets for halitosis prediction. Among the variable sets, Set 4 (basic information + oral diseases) consistently demonstrated the highest predictive performance across all ML models, suggesting that oral disease indicators (stomatitis, candidiasis, and periodontitis) are the most informative features for halitosis classification, regardless of the model architecture. In contrast, Set 1 (saliva-only data) and Set 2 (basic demographics + salivary flow + xerostomia) exhibited lower AUC scores across all models (0.60–0.62), indicating limited predictive ability when salivary or basic demographic information is used in isolation. Interestingly, the addition of systemic disease features (Set 5) or medication use data (Set 6) led to moderate improvement in the AUC in some models (Random Forest: 0.735 with medications), but not consistently across all models or to the extent observed in Set 4.

3.8 Performance of NN Models in Halitosis Prediction

Table 3 summarizes the performances of the four deep learning models trained using varying combinations of oral and systemic health features for halitosis prediction. Across all models, Set 4 (basic information + oral diseases) yielded the highest AUC, consistent with the findings for conventional ML models (Table 2). Notably, FT-Transformer achieved the highest predictive performance among all NN models, with an AUC of 0.761 (95% CI: 0.662–0.853), followed closely by SAINT with an AUC of 0.752 (95% CI: 0.651–0.840). In contrast, TabNet showed a relatively limited performance, particularly for Set 4 (AUC = 0.590), suggesting that attention-based architectures are more effective than sparse feature-selection models in this setting.

3.9 Performance Trend Analysis Across Feature Sets (Set 1–Set 7)

A comparison of feature sets and modeling approaches revealed key trends (Figure 7). Set 4 (basic information + oral diseases) consistently produced the highest AUC scores, emphasizing the importance of oral health in the prediction of halitosis. Specifically, Random Forest achieved an AUC of 0.760 (95% CI: 0.676–0.847), FT-Transformer scored 0.761 (95% CI: 0.662–0.853), and SAINT scored 0.752 (95% CI: 0.651–0.840), highlighting the significant contribution of oral disease features to systemic health or medication data. Adding more variables, such as systemic conditions (Set 5), medications (Set 6), or the full set of variables (Set 7), did not improve the performance and, in some cases, led to a slight decline. For example, the Random Forest AUC decreased from 0.760 (Set 4) to 0.680 (Set 7), and that of SAINT declined from 0.752 to 0.684. TabNet, the weakest neural model, showed a sharp decrease to 0.456 in Set 6, which is the lowest AUC across all models and feature sets. These results suggest that simply adding more features does not necessarily improve performance, possibly because of overfitting or the inclusion of less informative variables. When comparing deep learning models to traditional ML methods, FT-Transformer and SAINT outperformed Random Forest, whereas TabNet consistently underperformed. Ensemble methods combining ML and NN models did not show substantial improvements, with the ensemble AUC for Set 4 being 0.730, which is lower than that of FT-Transformer and Random Forest. This suggests that ensemble averaging may dilute the strengths of high-performing models, particularly when the base models have different generalization behaviors. In summary, oral disease features were the most important predictors of halitosis, and FT-Transformer and Random Forest were the most reliable models. Ensemble methods showed marginal improvements, underscoring the importance of thoughtful feature selection in clinical ML applications.

3.10 Set 2 vs. Set 3 AUC Comparison Across Models

To assess the effect of incorporating oral health features, we compared Set 2 (basic information) with Set 3 (basic + oral health characteristics) across all the ML and NN models. A model-wise p-value comparison (Table 4) showed that the performance of FT-Transformer significantly improved from Set 2 to Set 3 ($p = 0.049$), indicating that the addition of oral health features enhanced its predictive performance. In contrast, the four ML models (XGBoost, $p = 0.116$; CatBoost, $p = 0.361$) did not show significant improvements with the addition of oral health variables. This highlights the greater sensitivity of FT-Transformer to oral health features.

3.11 Comparative Evaluation of Model Performance Using Set 3 Features

To systematically evaluate the predictive performance of each model, we conducted the DeLong test to compare the AUC values derived from Set 3 (basic information + oral health characteristics) across eight models (four ML and four NN architectures) (Table 5). Among these, TabNet demonstrated the lowest AUC performance in both absolute and comparative terms. Table 5 summarizes the statistical comparisons between each model and TabNet.

Notably, two NN models, SAINT and FT-Transformer, showed a significantly higher predictive performance than TabNet, with p-values of 0.026 and 0.027, respectively. These differences were statistically significant at a threshold of 0.05, suggesting that attention-based deep learning models such as SAINT and FT-Transformer are better at utilizing oral health features for halitosis prediction. Other models, including TabTransformer, XGBoost, CatBoost, SVM, and Random Forest, had higher AUCs than TabNet; however, the difference was not statistically significant ($p > 0.05$). This indicates that while conventional ML models and other deep learning architectures perform reasonably well, SAINT and FT-Transformer offer a distinct advantage, possibly because of their superior ability to model complex feature interactions and non-linear dependencies in oral-systemic health data.

3.12 Feature Interaction, Correlation, and Importance in Halitosis Prediction

Figure 4 presents a comprehensive interpretability analysis of halitosis prediction using a transformer-based NN model that incorporates attention-derived feature interactions, point-biserial correlations with the target variable, and permutation-based importance rankings. Together, these analyses shed light on the direct and pairwise influences of individual variables across the salivary, demographic, and oral health domains.

3.12.1 Top 10 Feature Pairs with Highest Attention Scores

Panel A presents the top 10 feature pairs with the highest attention scores, excluding self-interactions. These scores are derived from the internal attention mechanism of the transformer model, which quantifies the extent to which the model emphasizes the interaction between two features during prediction. An attention score approaching 1.0 reflects a strong model-learned dependency, indicating that the feature pair is highly informative for the prediction task (i.e., halitosis prediction). Conversely, lower scores suggest weaker or less influential interactions as perceived by the model.

Among the top-ranked pairs, BufferCapacity $\leftrightarrow$ UFR (attention score = 0.998), sticky saliva $\leftrightarrow$ UFR (0.994), and O_Glossodynia $\leftrightarrow$ UFR (0.989) exhibited the strongest interactions. These results suggest that the UFR consistently plays a central role in pairwise feature dynamics and can act as an important moderator of the effects of other variables. The pattern of interactions emphasizes that the salivary function status significantly interacts with oral symptom-related variables, such as mucosal pain and hygiene, in the prediction of halitosis.

3.12.2 Feature–Target Correlation

Panel B displays the point-biserial correlations between individual features and the binary halitosis label. The strongest positive correlations were observed for O_Stomatitis ($r = 0.079$) and Xerostomia_subjective ($r = 0.073$), suggesting that these features were more frequently associated with halitosis. In contrast, SFR ($r = -0.149$), age ($r = -0.114$), and O_RAU ($r = -0.086$) were negatively correlated with halitosis, implying that lower salivary flow rate and younger age may be linked to greater halitosis risk. However, the overall correlation coefficients were modest (range, -0.15 to $+0.08$), indicating that linear relationships alone are

insufficient to explain the outcome variance.

3.12.3 Permutation Importance Ranking

Panel C presents the permutation importance rankings, highlighting the mean drop in the AUC upon feature shuffling. Age (mean $\Delta\text{AUC} = 0.094$), O_Stomatitis (mean $\Delta\text{AUC} = 0.091$), and UFR (mean $\Delta\text{AUC} = 0.068$) were the three most important features in the model. These results reinforce the clinical relevance of both salivary quantity and mucosal condition in predicting halitosis. Further, O_BMS (mean $\Delta\text{AUC} = 0.035$), O_Candidiasis (mean $\Delta\text{AUC} = 0.030$), and O_RAU (mean $\Delta\text{AUC} = 0.027$) exhibited moderate contributions, whereas Xerostomia_subjective, BufferCapacity, and SFR had negligible or even negative impact on model performance when permuted.

3.13 Age-Stratified Permutation Feature Importance Across ML Models

To better understand how feature relevance varies across age groups, permutation importance was calculated separately within three age groups: <30 years (Panel A), 30–49 years (Panel B), and ≥ 50 years (Panel C). Permutation importance was quantified as ΔAUC , the difference in model performance AUC before and after random shuffling of a given feature. Specifically, $\Delta\text{AUC} = \text{AUC}_{\text{original}} - \text{AUC}_{\text{permuted}}$. A positive ΔAUC indicates that permuting the feature reduces model performance, suggesting that the feature contributes meaningfully to the prediction. In contrast, a negative ΔAUC implies that the feature may introduce noise or interfere with model generalization, as its removal actually improves performance. In the youngest age group (<30 years), salivary function features, such as SFR, UFR, pH, and BufferCapacity, showed the highest importance. Notably, BufferCapacity had a strongly negative ΔAUC in XGBoost (-0.233), indicating potential model overfitting or an unreliable pattern learned from the data. Oral diseases, such as O_Stomatitis and O_RAU, demonstrated moderate positive importance in CatBoost and SVM (O_Stomatitis = 0.111 in CatBoost; O_RAU = 0.122 in SVM). In contrast, female sex and VAS scores negatively impacted model performance in SVM (-0.189 and -0.122 , respectively), suggesting a potential confounding or unstable contribution in that model. No single dominant feature emerged among participants aged 30–49 years. Most ΔAUC values were small and scattered across features and models. Calculus showed a negative importance in XGBoost (-0.094), and O_RAU showed moderate importance only in SVM (0.058). This diffuse pattern suggests reduced predictive clarity in this age group, possibly because of greater heterogeneity or weaker associations with halitosis. In the older group (≥ 50 years), subjective xerostomia, UFR, SFR, and VAS scores were the most predictive features. For example, Xerostomia_subjective and SFR had positive ΔAUC values in XGBoost (0.031 and 0.027, respectively), and VAS score exhibited a ΔAUC value of 0.035. Systemic health indicators such as number of systemic diseases and S_Hypertension showed small but consistent contributions ($\Delta\text{AUC} \approx 0.01$). Interestingly, O_LichenPlanus negatively impacted performance in SVM (-0.027), perhaps due to low prevalence or feature redundancy in older individuals.

3.14 Age-Stratified SHAP Analysis of Halitosis Prediction Models

Figure 6 presents SHAP (SHapley Additive exPlanations) summary plots for age-stratified model interpretation, comparing XGBoost (Panels A–C) and CatBoost (Panels D–F) across three age groups: <30 years, 30–49 years, and ≥ 50 years. In the youngest group (<30 years), SFR, pH, VAS score, and buffering capacity were the key contributors in both XGBoost (Panel A) and CatBoost (Panel D). High SFR and pH values generally reduced the predicted risk of halitosis, whereas low values of buffering capacity and oral hygiene were associated with an increased risk. Notably, oral stomatitis and age showed a moderate impact, suggesting that early mucosal conditions and age-related factors may influence model decisions even in younger individuals. For the middle-age group (30–49 years), the feature influence was more diffuse. In XGBoost (Panel E), calculus, pH, and SFR were the most significant features, albeit with mixed directionality. Oral hygiene, UFR, and subjective xerostomia were moderate contributors in CatBoost (Panel B). These findings suggest a lack of dominant predictive drivers in this age group, consistent with the weaker permutation importance patterns previously observed. Among older adults (≥ 50 years), stronger and more clinically intuitive patterns were observed in both models. Oral stomatitis, UFR, age, buffering capacity, and systemic diseases (hypertension and CVD) played major roles in XGBoost (Panel C). SHAP values revealed that a lower UFR and older age increased halitosis risk, whereas higher salivary buffering capacity values were protective. Oral stomatitis, buffer capacity, and S_Hypertension emerged as key features in CatBoost (Panel F), confirming the relevance of both local (oral) and systemic indicators in late adulthood.

4. Discussion

In this study, we identified the clinical and systemic factors associated with self-reported halitosis using conventional ML and transformer-based deep learning models. We also compared the individual models and ensemble approaches to determine the most effective predictive strategy. Contrary to our hypothesis, expanding the input variables, including salivary flow rates, oral and systemic health conditions, and medication use, did not improve model performance. Instead, models trained on selected features, particularly oral disease indicators, consistently outperformed those trained using the full variable set. This suggests that including less informative variables may introduce noise, reduce generalizability, and increase overfitting risk [26]. Moreover, ensemble methods combining high-performance models such as FT-Transformer and Random Forest yield no substantial performance gains, highlighting the importance of model parsimony and thoughtful feature selection over brute-force inclusion [27]. Finally, explainable AI techniques, such as SHAP analysis, attention-based feature interaction, and permutation importance, provided clinically meaningful insights and clarified age-specific patterns, enhancing the translational value of our findings.

Among all the predictors, salivary function, particularly UFR and buffering capacity, consistently demonstrated the strongest association with halitosis across model architectures and age strata. In addition, SFR was significantly lower in the halitosis group than in the non-halitosis group. This aligns with well-established pathophysiological mechanisms wherein decreased salivary flow and impaired buffering capacity have been reported to create an acidic and anaerobic oral environment, thereby facilitating the growth of VSC-producing bacteria such as *Prevotella*, *Fusobacterium*, and *Porphyromonas* species [28, 29]. Notably, UFR showed the highest feature importance in both younger (<30) and older (≥ 50) adults. Patients with a reduced UFR were significantly more likely to report oral dryness and halitosis symptoms [16]. Similarly, Narhi (1994) reported that hyposalivation, especially during the resting state, is associated with a higher microbial load and halitosis, even when SFRs remain normal [30]. Furthermore, a lower UFR correlates with a higher plaque index, gingival inflammation, and tongue coating, which contribute to intraoral halitosis [31]. In our study, the buffering capacity also exhibited a strong negative influence, especially among younger adults, indicating that even minor reductions in this protective function may result in perceived halitosis. Low salivary buffering capacity correlates with increased acidity in the oral cavity and greater VSC concentration [32]. Taken together, these results reinforce the notion that UFR and buffering capacity are not only objective indicators of oral dryness and host defense, but also strong subjective predictors of halitosis, supporting their inclusion in both clinical evaluation and AI-driven diagnostic models.

Oral mucosal conditions such as stomatitis, BMS, and glossodynia were key predictors of halitosis in older adults, with SHAP analysis showing significant contributions in the ≥ 50 age group. Chronic inflammation or neuropathic pain in the oral mucosa may increase

somatosensory sensitivity, leading to a greater awareness of minor odors or discomfort [33, 34]. In patients with BMS and glossodynia, peripheral nerve sensitization and altered central pain processing have been associated with increased reports of oral symptoms, including halitosis [35, 36]. Stomatitis, which is often associated with xerostomia, denture irritation, or immune dysregulation, promotes the dysbiosis of the oral microbiome [37]. Inflammatory mucosal environments tend to favor the growth of anaerobic bacteria, such as *Fusobacterium nucleatum* and *Prevotella intermedia*, which are known producers of VSCs [38, 39]. This shift may elevate both actual odor levels and subjective halitosis perception. As this study did not assess the oral microbiome, further research is warranted to elucidate the mechanisms underlying these associations.

Systemic health conditions, including hypertension, osteoporosis, and the cumulative number of chronic diseases, were included in the predictive models but exhibited only modest to weak associations with self-reported halitosis. Although their contribution was consistently lower than that of oral factors, these variables may indirectly influence halitosis, primarily through mechanisms such as medication-induced xerostomia, reduced immune function, or altered salivary gland physiology. Several systemic diseases affect the salivary flow and composition. For example, patients with hypertension or DM often use medications such as antihypertensive or antidiabetic drugs, which impair salivary secretion or alter oral pH, thereby creating an environment conducive to VSC production [40, 41]. Polypharmacy, which is common in older adults, further increases the risk of salivary gland hypofunction and subjective oral dryness, which are risk factors for halitosis [42]. However, in our models, the effect sizes for the systemic variables were relatively small, and no single systemic disease emerged as a dominant predictor of halitosis. This finding is consistent with previous studies indicating that systemic health alone is rarely sufficient to explain subjective halitosis without concomitant oral dysfunction [43, 44]. Rather, systemic conditions may act as modifiers, exacerbating pre-existing oral issues, or lowering the threshold for the perception of halitosis symptoms. These observations suggest that, while systemic health should not be overlooked, its diagnostic value in halitosis prediction is limited unless evaluated in conjunction with oral and salivary parameters. A comprehensive clinical approach that integrates both local and systemic assessments is essential for the accurate evaluation of halitosis.

Feature contributions varied substantially across the age groups, suggesting age-specific mechanisms underlying self-perceived halitosis. In younger individuals (age <30 years), the salivary flow rate and buffering capacity were consistently the most influential predictors, emphasizing the role of salivary gland function in early adulthood when the systemic disease burden is low and awareness of oral symptoms is high [45]. In contrast, the ≥50-year age group showed stronger associations with mucosal and neuropathic conditions such as stomatitis, BMS, and glossodynia, possibly because of the increased prevalence of chronic mucosal disease, polypharmacy, and salivary gland atrophy [46]. The shift in the importance of predictors with age reflects a broader transition from physiological to pathological influences on oral health. Interestingly, the middle-age group (30–49 years) demonstrated a lower overall feature importance and more diffuse pattern, possibly due to unmeasured factors such as psychosocial

stress, hormonal changes, or behavioral habits (smoking and alcohol use) [13, 43]. Over one-third of individuals aged ≥ 60 years report subjective halitosis [47], and salivary quantity, composition, and function are known to decline with age [48, 49]. These findings support age-tailored screening and intervention strategies, with salivary function prioritized in younger adults, mucosal pathology in older individuals, and broader assessments in middle-aged populations. Incorporating age-specific models or feature sets into AI-based clinical tools may enhance diagnostic precision and relevance.

The application of explainable AI techniques, such as SHAP analysis, permutation importance, and attention-based interpretation, provide valuable insights into the decision-making processes of complex models. These tools enable the identification of clinically relevant features, age-specific predictor patterns, and non-linear interactions that are often missed by traditional statistics [50, 51]. Among the models, FT-Transformer consistently outperformed the others, particularly when incorporating oral health variables such as buffering capacity and mucosal symptoms. Its ability to model complex feature interactions possibly explains its superior predictive accuracy over conventional models, such as logistic regression or Random Forest. The model also showed marked performance gains with expanded feature sets, underscoring its capacity to extract additional clinical value from nuanced data [52]. From a clinical perspective, these results highlight the importance of interpretability in oral health research. Although black-box predictions may be accurate, they require transparent reasoning to be trusted and actionable [53]. Explainable AI enables clinicians to visualize the impact of specific factors, justify model decisions, and communicate risks to patients in a more effective manner. In cases of subjective halitosis, where objective findings may be subtle, such tools help bridge the gap between patient complaints and clinical evidence. These methods also support precise oral health assessment by enabling age- and context-specific modeling, thereby improving risk stratification and guiding personalized care.

This study had several notable strengths. First, it used a relatively large and clinically diverse sample ($N = 821$), enabling a stratified analysis across different age groups and health profiles. Second, the integration of both oral and systemic variables allowed for a multifactorial understanding of halitosis, extending beyond conventional dental indicators. Third, to address group imbalance in the dataset, particularly the small proportion of participants reporting halitosis, we applied SMOTE. This method improves the reliability and stability of models by mitigating the risk of biased learning in underrepresented classes [54]. Fourth, this study employed a variety of predictive models, including transformer-based deep learning models, and applied explainable AI techniques, such as SHAP, permutation importance, and attention-based interpretation, to enhance the interpretability and clinical applicability of the results.

Nevertheless, this study has some limitations. The assessment of halitosis relied solely on self-reporting, without objective measurements such as organoleptic scores or gas chromatography, which may introduce bias stemming from individual perception, psychological sensitivity, or social context. Second, data pertaining to important behavioral and psychosocial variables, including stress, smoking, alcohol intake, and dietary habits, were not available, limiting model performance, particularly in the middle-age group, where etiologies may be more

heterogeneous. Third, the study was conducted at a single clinical center, which may restrict the generalizability of the findings to broader populations with different demographic and cultural backgrounds. Fourth, the oral microbiome and tongue coating, both major contributors to intraoral halitosis, were not assessed, potentially leaving key mechanistic pathways unexplored. Future research should incorporate objective diagnostic tools and broaden variable inclusion to encompass behavioral, microbial, and psychosocial dimensions. External validation across multicenter cohorts and longitudinal follow-ups would further enhance model robustness and clarify the causal relationships between predictive factors and halitosis symptoms.

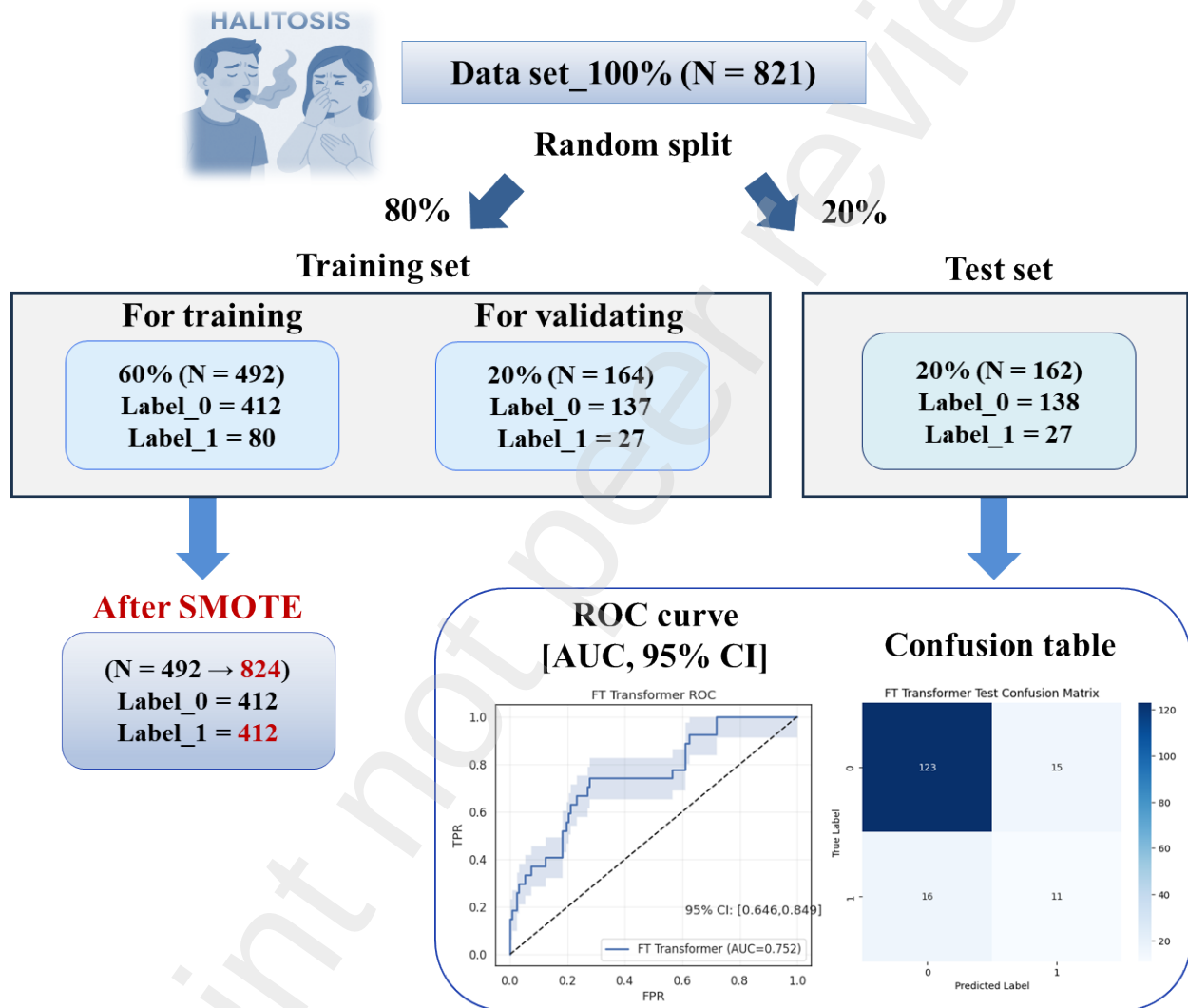
Conclusion

This study highlights the utility of ML and transformer-based deep learning models in identifying key clinical and systemic factors associated with self-reported halitosis. Salivary dysfunction, particularly reduced UFR and impaired buffering capacity, emerged as the most consistent and influential predictor in all age groups. Oral mucosal conditions such as stomatitis, BMS, and glossodynia played a particularly important role in older adults, whereas systemic diseases showed only modest associations. The importance of individual predictors varied significantly with age, highlighting the need for tailored assessment strategies. The integration of explainable AI techniques enabled the transparent interpretation of model outputs and revealed nuanced age-specific feature interactions that may not be captured through conventional analysis. Transformer-based models, particularly FT-Transformer, demonstrated superior performance by effectively leveraging complex feature interactions and additive clinical information. These findings support the development of precise oral health tools that can accommodate the multifactorial age-dependent nature of halitosis. To improve future predictive models, the incorporation of objective diagnostic tools and additional behavioral, psychosocial, and microbiome-related variables is warranted. Moreover, adopting explainable AI frameworks in clinical practice may enhance diagnostic precision, improve communication with patients, and guide personalized, context-sensitive interventions for managing halitosis.

Figures and Tables

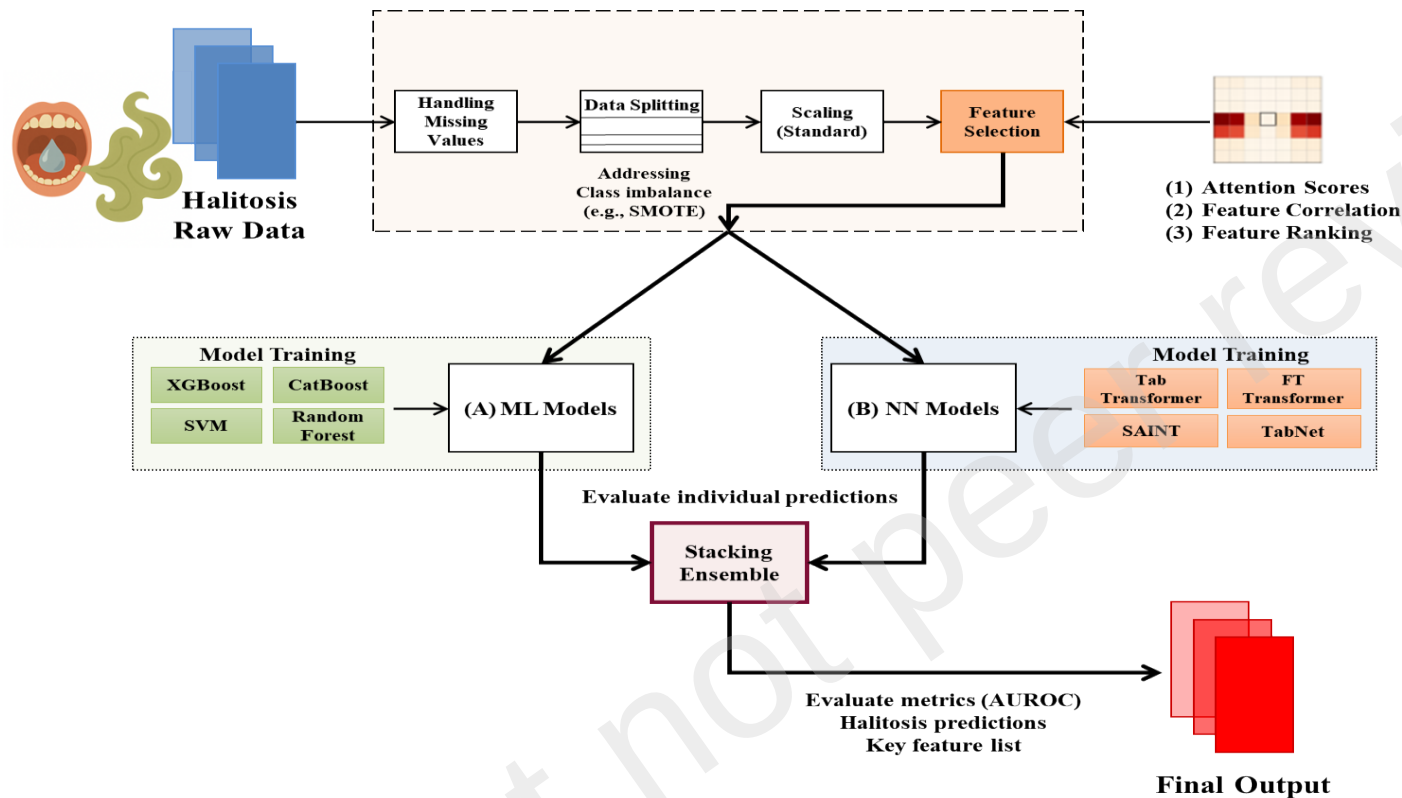
Figure 1. Data preprocessing, resampling, and model evaluation pipeline.

The dataset (N = 821) was divided into training (60%), validation (20%), and testing (20%) subsets. To address the class imbalance, the training set was oversampled to ensure equal representation of both classes. Model evaluation was conducted using methods such as the ROC analysis and confusion matrix for the testing set, as illustrated in the example.



N: number of patients, Label_0: Negative class (Non-halitosis), Label_1: Positive class (Halitosis), SMOTE: Synthetic Minority Over-sampling Technique, ROC: Receiver Operating Characteristic, AUC: Area Under the Curve, CI: Confidence Interval, TPR: True Positive Rate, FPR: False Positive Rate.

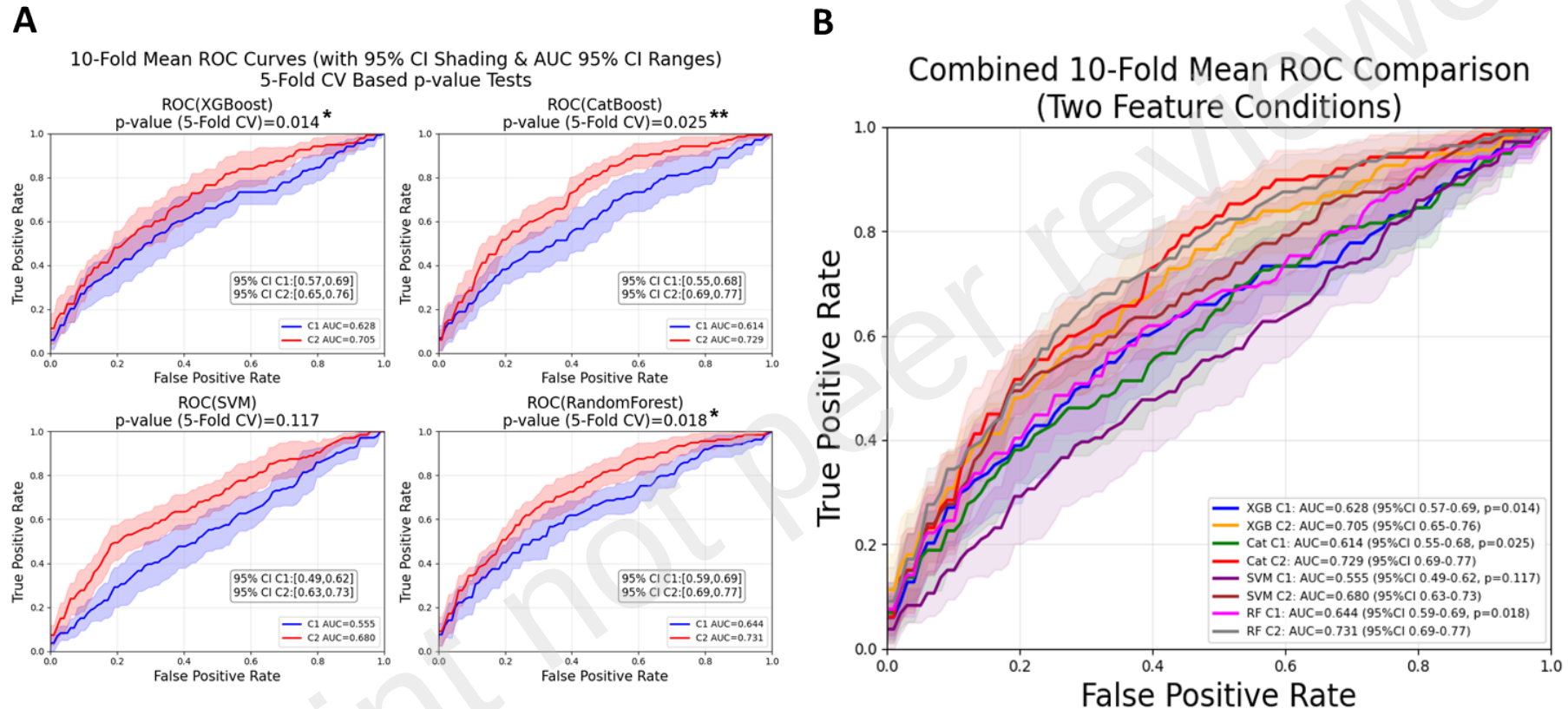
Figure 2. Schematic overview of the halitosis prediction framework.



The framework consists of data preprocessing steps, including missing value imputation, class imbalance correction (e.g., SMOTE), standard scaling, and feature selection based on attention scores, feature correlations, and permutation importance rankings. Subsequently, two model families were trained: (A) conventional machine learning models and (B) four neural-network-based transformer models. Individual model predictions were aggregated through a stacking ensemble for final halitosis prediction and extracting key predictive features, which were evaluated using metrics such as AUC.

XGBoost: eXtreme Gradient Boosting, CatBoost: Categorical Boosting, SVM: Support Vector Machine, Random Forest: Random Forest Classifier, NN: Neural Network, TabTransformer: Tabular Transformer, FT-Transformer: Feature Tokenizer Transformer, SAINT: Self-Attention and Intersample Attention Transformer, TabNet: Tabular Network (Deep learning architecture for tabular data using attentive feature selection), SMOTE: Synthetic Minority Over-sampling Technique, Scaling (Standard): Standardization of input features (usually to zero mean and unit variance), ROC: Receiver Operating Characteristic, AUC: Area Under the Curve.

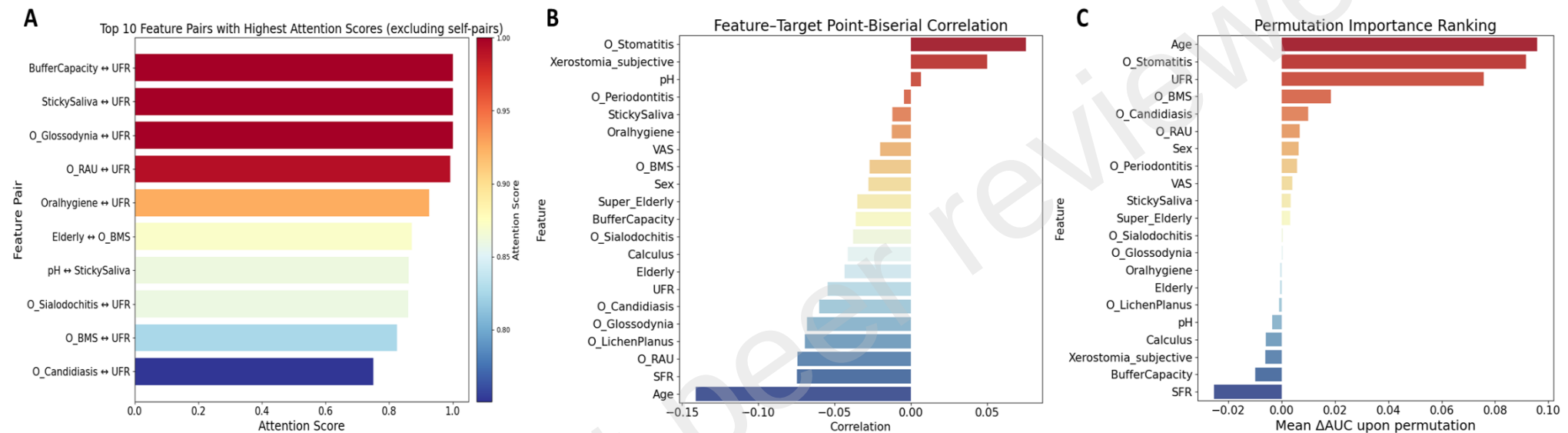
Figure 3. Comparative ROC analysis of machine learning models under two feature conditions for halitosis prediction.



(A) Ten-fold mean ROC curves and AUC comparisons across models with 95% confidence intervals.
(B) The combined ROC curve overlay highlighting performance differences between feature conditions across all models.

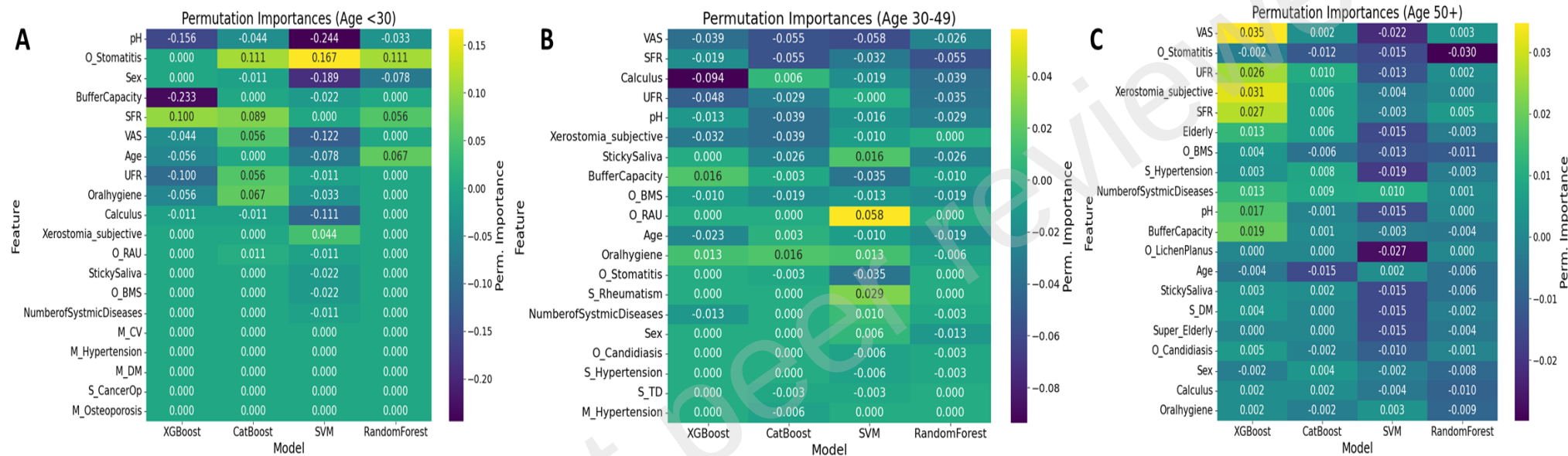
ROC, receiver operating characteristic; CI, confidence interval; AUC, area under the curve; CV, cross-validation; C1, Condition 1 (basic information + oral health characteristics); C2, Condition 2 (basic information + oral diseases); SVM, support vector machine; XGBoost (XGB), extreme gradient boosting; CatBoost (Cat), categorical boosting; Random Forest (RF), random forest classifier.

Figure 4. Attention-based feature interaction, correlation, and importance analysis in halitosis prediction.



This figure provides a comprehensive analysis of the feature relationships and their contributions to model performance using a transformer-based attention mechanism and correlation analysis. It includes the attention scores between feature pairs (Panel A), point-biserial correlations with the target variable (Panel B), and individual feature importance rankings based on permutation analysis (Panel C).

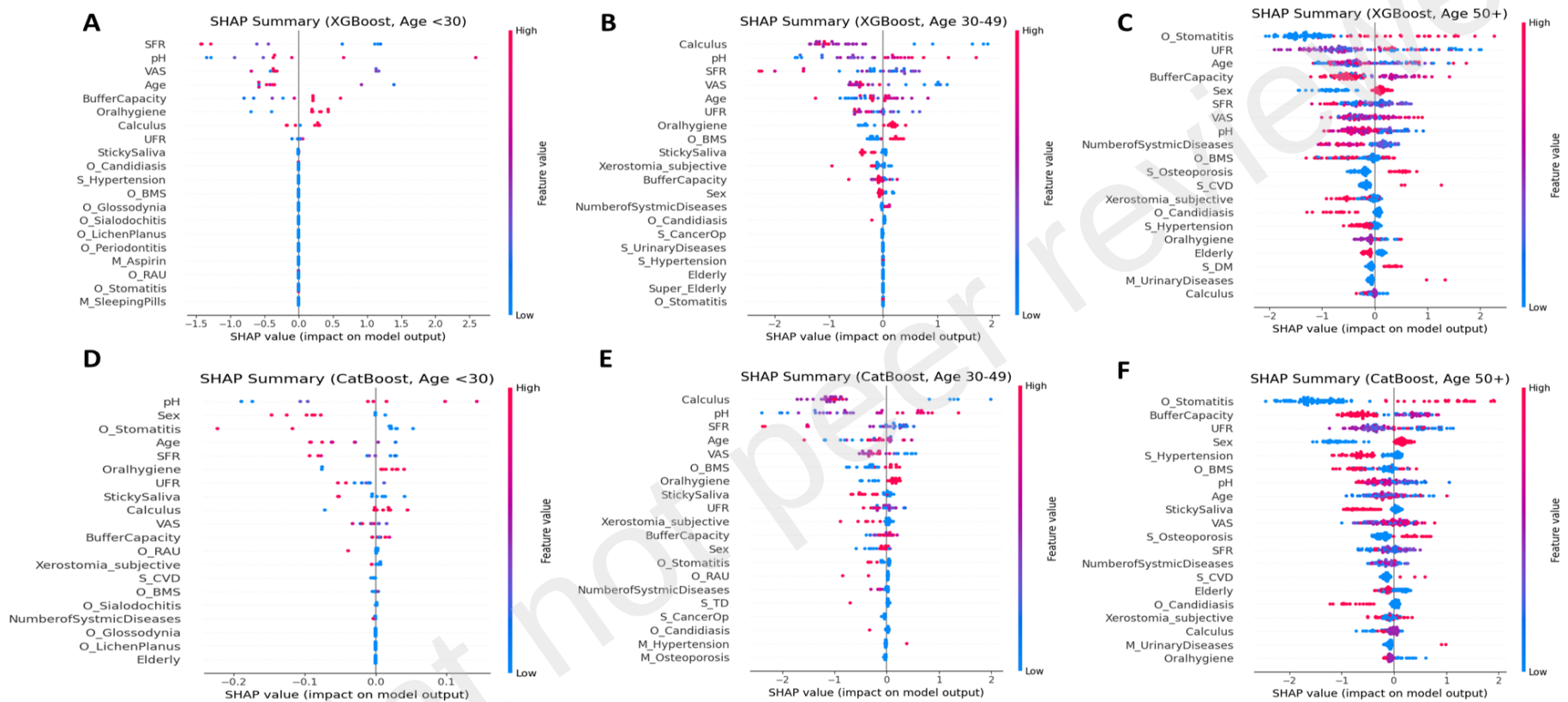
Figure 5. Age-stratified permutation feature importance across ML models for halitosis prediction



UFR: Unstimulated Flow Rate, SFR: Stimulated Flow Rate, pH: Salivary pH. BufferCapacity: Salivary Buffering Capacity, Xerostomia_subjective: Subjective Dry Mouth Sensation, StickySaliva: Presence of Sticky Saliva, Oralhygiene: Oral Hygiene Status, Calculus: Calculus Deposition, VAS: Visual Analogue Scale for Oral Pain, O_Stomatitis: Oral Stomatitis, O_RAU: Recurrent Aphthous Ulcers, O_Candidiasis: Oral Candidiasis, O_Periodontitis: Periodontitis, O_LichenPlanus: Oral Lichen Planus, O_Sialodochitis: Sialodochitis, O_Glossodynia: Glossodynia, O_BMS: Burning Mouth Syndrome, S_Hypertension: Systemic Hypertension, S_DM: Diabetes Mellitus, S_Hyperlipidemia: Hyperlipidemia, S_Renal diseases: Renal Diseases, S_Liver diseases: Liver Diseases, S_Rheumatism: Rheumatic Diseases, S_Osteoporosis: Osteoporosis, S_CVD: Cardiovascular Disease, S_TD: Thyroid Disease, S_Mental disorders: Mental Disorders, S_Urinary diseases: Urinary Diseases, S_Asthma: Asthma, S_Cancer Op: History of Cancer Operation, M_Hypertension: Medication for Hypertension, M_CV: Medication for Cardiovascular Disease, M_DM: Medication for Diabetes Mellitus, M_Osteoporosis: Medication for Osteoporosis, Elderly: Age ≥ 65 , Super_Elderly: Age ≥ 75

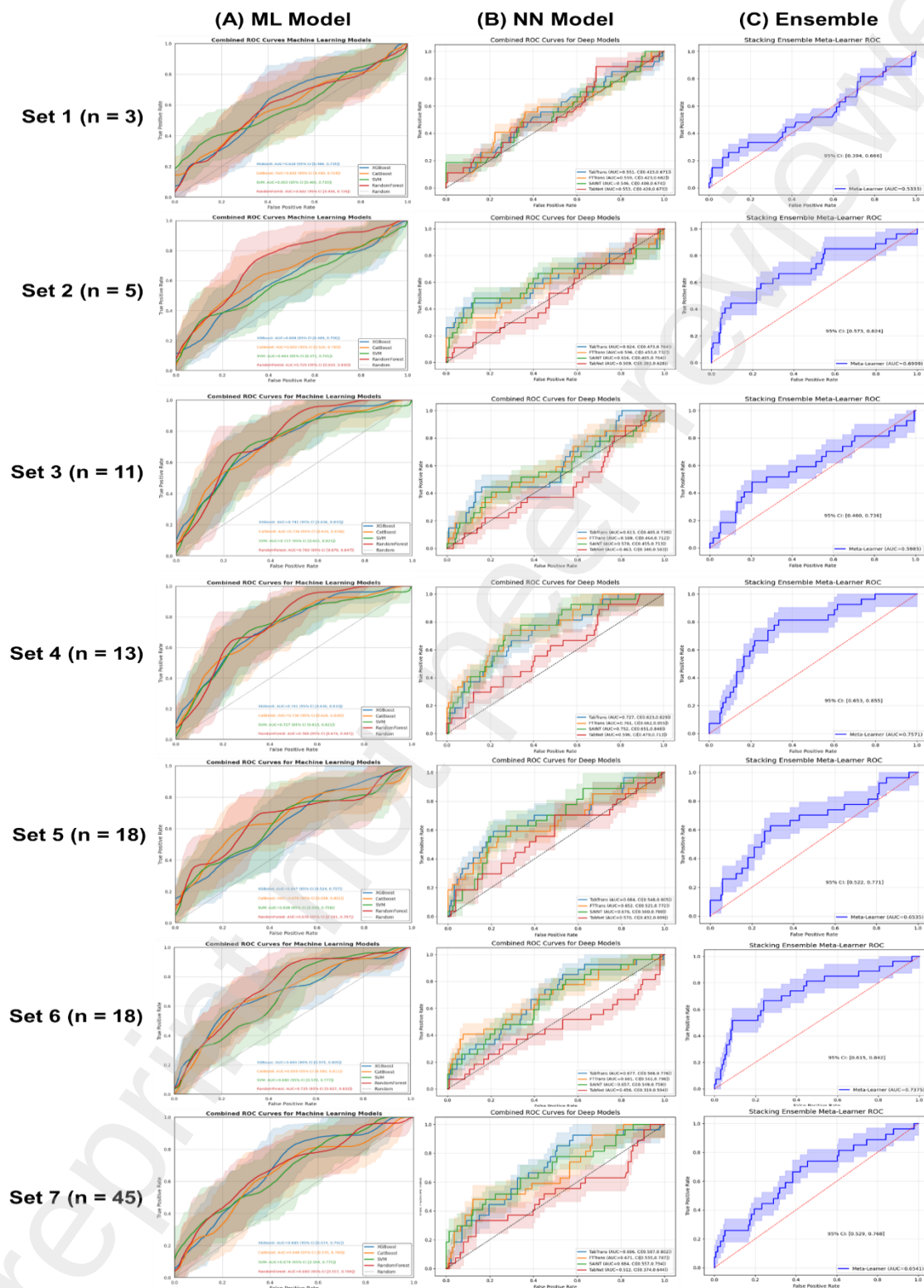
Panels A–C show the permutation importance scores for participants aged <30, 30–49, and ≥ 50 years, respectively, across four models (XGBoost, CatBoost, SVM, and Random Forest). Values represent ΔAUC (i.e., the change in AUC upon permutation), where positive scores indicate informative features (greater model dependence) and negative scores indicate potentially noisy or misleading features. Salivary function-related features (UFR, SFR, and pH) and mucosal conditions (stomatitis) were key predictors in the younger and older groups, with variations in feature importance by age group and model architecture.

Figure 6. Age-stratified SHAP analysis for halitosis prediction using XGBoost and CatBoost models.



This figure presents SHapley Additive exPlanations (SHAP) summary plots, highlighting the relative contribution and directionality of each feature to model output across different age groups. The top three plots (A–C) represent the XGBoost model and the bottom three plots (D–F) represent the CatBoost model. The colors represent the actual feature values (blue: low; red: high), and the horizontal position indicates the impact on prediction by the model (positive or negative influence).

Figure 7. ROC analysis of machine-learning and transformer models across seven feature sets.



Set 1: Saliva – Salivary Flow Rate and Xerostomia (includes: unstimulated flow rate (UFR), stimulated flow rate (SFR), and subjective xerostomia); Set 2: Basic Information (includes: sex, age, UFR, SFR, and subjective xerostomia); Set 3: Basic + Oral Health (includes all variables in Set 2 plus: salivary pH, buffer capacity, visual

analog scale (VAS) for oral pain, sticky saliva, oral hygiene, and calculus deposition); Set 4: Basic + Oral Diseases (includes all variables in Set 2 plus oral disease indicators: O_Stomatitis, O_RAU, O_Candidiasis, O_Periodontitis, O_LichenPlanus, O_Sialodochitis, O_Glossodynia, and O_BMS); Set 5: Basic + Systemic Diseases (includes all variables in Set 2 plus systemic diseases: S_Hypertension, S_DM, S_Hyperlipidemia, S_Renal diseases, S_Liver diseases, S_Rheumatism, S_Osteoporosis, S_CVD, S_TD, S_Mental disorders, S_Urinary diseases, S_Asthma, and S_Cancer Op); Set 6: Basic + Medications (includes all variables in Set 2 plus systemic medications: M_Hypertension, M_DM, M_Osteoporosis, M_Hyperlipidemia, M_CV, M_TD, M_GI, M_Urinary diseases, M_Arthritis, M_Rheumatism, M_Anxiolytic, M_Sleeping pills, and M_Aspirin); Set 7: All Variables (includes all variables from Sets 1 through 6).

The rows correspond to progressively expanded variable sets, from the baseline demographic-salivary panel (row 1) to the full feature set (row 6). Column A shows the conventional ML models, column B shows the transformer-based neural networks, and column C shows the stacking-ensemble meta-learner. Solid lines denote mean ROC curves; shaded ribbons indicate the bootstrapped 95% confidence interval of the true-positive rate. FT-Transformer consistently achieved the highest AUC (column B), with a statistically significant superiority over TabNet (Table 5). Ensemble stacking did not yield additional benefits beyond those of the strongest individual learners.

Table 1. Distribution of demographics and oral health factors according to halitosis status

Variable	Non-Halitosis (n=687) Mean ± SD or n (%)	Halitosis (n=134) Mean ± SD or n (%)	p-value
(1) Basic information			
Age (years) ^a	60.38 ± 15.41	54.12 ± 20.14	0.0001***
Age group 1 (<30 years) ^b	29 (4.2%)	16 (11.9%)	0.0011**
Age group 2 (30-59 years) ^b	277 (40.3%)	55 (41.0%)	
Age group 3 (>60 years) ^b	381 (55.5%)	63 (47.0%)	
Sex (Female) ^c	495 (84.3%)	92 (15.7%)	0.4890
UFR (mL/min) ^a	0.39 ± 0.24	0.35 ± 0.26	0.1182
SFR (mL/min) ^a	1.32 ± 0.72	1.17 ± 0.56	0.0321*
Subjective xerostomia ^c	160 (23.3%)	39 (29.1%)	0.1846
(2) Oral health characteristics			
Salivary pH ^a	6.72 ± 0.68	6.73 ± 0.63	0.8487
Buffer Capacity ^a	10.58 ± 2.23	10.36 ± 2.37	0.3007
VAS for oral pain ^a	5.62 ± 2.97	5.46 ± 3.32	0.5650
Sticky saliva ^c	200 (29.1%)	37 (27.6%)	0.8054
Oral hygiene [Good] ^b	125 (18.2%)	26 (19.4%)	0.9356
Oral hygiene [Acceptable] ^b	486 (70.7%)	94 (70.1%)	
Oral hygiene [Poor] ^b	76 (11.1%)	14 (10.4%)	
Calculus deposition [None] ^b	135 (19.7%)	32 (23.9%)	0.4879
Calculus deposition [Moderate] ^b	501 (72.9%)	94 (70.1%)	
Calculus deposition [Severe] ^b	51 (7.4%)	8 (6.0%)	
(3) Oral Disease			
O_Stomatitis ^c	124 (18.0%)	35 (26.1%)	0.0411*
O_RAU ^d	41 (6.0%)	2 (1.5%)	0.0555
O_Candidiasis ^c	106 (15.4%)	13 (9.7%)	0.1121
O_Periodontitis ^d	22 (3.2%)	4 (3.0%)	1.0000

O_Lichen Planus ^d	38 (5.5%)	2 (1.5%)	0.0772
O_Sialodochitis ^d	6 (0.9%)	0 (0.0%)	0.5951
O_Glossodynia ^d	19 (2.8%)	0 (0.0%)	0.1023
O_BMS ^c	234 (34.1%)	41 (30.6%)	0.4983

UFR: Unstimulated salivary flow rate; SFR: Stimulated salivary flow rate; RAU: Recurrent aphthous ulcers; BMS: Burning mouth syndrome; O_: Prefix indicating oral diseases; SD: Standard deviation.

Values are presented as mean $\pm$ SD for continuous variables and as number (percentage) for categorical variables.

^a: p-values for continuous variables were calculated using the independent t-test with Bonferroni correction. ^b: p-values for categorical variables were calculated using the chi-square test with Bonferroni correction (2 $\times$ 2 tables).

^c: p-values for categorical variables were calculated using the chi-square test with Bonferroni correction (2 $\times$ 3 tables). ^d: p-values for categorical variables were calculated using Fisher's exact test. A p-value < 0.05 was considered statistically significant. ***p < 0.001; **p < 0.01; p* < 0.05.

Table 2. AUC performance of the four conventional machine learning models

Model	Variable Set	AUROC (95% CI)
XGBoost	Set 1: Saliva (n = 3)	0.618 (0.486-0.735)
	Set 2: Basic information (n = 5)	0.608 (0.480-0.730)
	Set 3: Basic information + Oral health (n = 11)	0.623 (0.496-0.753)
	Set 4: Basic information + Oral diseases (n = 13)	0.741 (0.636-0.833)
	Set 5: Basic information + Systemic diseases (n = 18)	0.647 (0.524-0.757)
	Set 6: Basic information + Medications (n = 18)	0.694 (0.575-0.805)
	Set 7: All variables (n = 45)	0.685 (0.574-0.792)
CatBoost	Set 1: Saliva	0.602 (0.460-0.726)
	Set 2: Basic information	0.659 (0.520-0.785)
	Set 3: Basic information + Oral health	0.643 (0.513-0.770)
	Set 4: Basic information + Oral diseases	0.736 (0.626-0.838)
	Set 5: Basic information + Systemic diseases	0.678 (0.539-0.801)
	Set 6 6: Basic information + Medications	0.699 (0.580-0.811)
	Set 7: All variables	0.648 (0.535-0.760)
SVM	Set 1: Saliva	0.603 (0.460-0.733)
	Set 2: Basic information	0.604 (0.471-0.741)
	Set 3: Basic information + Oral health	0.566 (0.461-0.676)
	Set 4: Basic information + Oral diseases	0.727 (0.615-0.821)
	Set 5: Basic information + Systemic diseases	0.638 (0.506-0.758)
	Set 6 6: Basic information + Medications	0.680 (0.578-0.777)
	Set 7: All variables	0.674 (0.564-0.775)
Random Forest	Set 1: Saliva	0.602 (0.458-0.726)
	Set 2: Basic information	0.729 (0.619-0.830)
	Set 3: Basic information + Oral health	0.661 (0.539-0.774)
	Set 4: Basic information + Oral diseases	0.760 (0.676-0.847)

	Set 5: Basic information + Systemic diseases	0.670 (0.541-0.797)
	Set 6 6: Basic information + Medications	0.735 (0.627-0.832)
	Set 7: All variables	0.680 (0.557-0.784)

XGBoost: Extreme Gradient Boosting, CatBoost: Categorical Boosting, SVM: Support Vector Machine, Random Forest: Random Forest Classifier.

Table 3. AUC performance of four deep neural network models using combined oral-systemic features

Model	Variable Set	AUROC (95% CI)
Tab Transformer	Set 1: Saliva (n = 3)	0.551 (0.423-0.671)
	Set 2: Basic information (n = 5)	0.624 (0.473-0.764)
	Set 3: Basic information + Oral health (n = 11)	0.613 (0.485-0.739)
	Set 4: Basic information + Oral diseases (n = 13)	0.727 (0.623-0.829)
	Set 5: Basic information + Systemic diseases (n = 18)	0.684 (0.548-0.805)
	Set 6: Basic information + Medications (n = 18)	0.677 (0.566-0.776)
	Set 7: All variables (n = 45)	0.691 (0.587-0.802)
FT Transformer	Set 1: Saliva	0.559 (0.423-0.682)
	Set 2: Basic information	0.596 (0.453-0.732)
	Set 3: Basic information + Oral health	0.588 (0.464-0.712)
	Set 4: Basic information + Oral diseases	0.761 (0.662-0.855)
	Set 5: Basic information + Systemic diseases	0.652 (0.521-0.772)
	Set 6: Basic information + Medications	0.681 (0.561-0.798)
	Set 7: All variables	0.671 (0.555-0.787)
SAINT	Set 1: Saliva	0.546 (0.408-0.674)
	Set 2: Basic information	0.616 (0.465-0.764)
	Set 3: Basic information + Oral health	0.578 (0.455-0.713)
	Set 4: Basic information + Oral diseases	0.752 (0.651-0.848)
	Set 5: Basic information + Systemic diseases	0.676 (0.560-0.780)
	Set 6: Basic information + Medications	0.657 (0.549-0.759)
	Set 7: All variables	0.684 (0.557-0.794)
TabNet	Set 1: Saliva	0.553 (0.428-0.673)
	Set 2: Basic information	0.509 (0.383-0.628)
	Set 3: Basic information + Oral health	0.463 (0.340-0.583)
	Set 4: Basic information + Oral diseases	0.596 (0.479-0.713)
	Set 5: Basic information + Systemic diseases	0.570 (0.560-0.780)
	Set 6: Basic information + Medications	0.456 (0.319-0.594)

Stacking Ensemble (XGBoost, Random Forest, Tab Transformer, FT Transformer)	Set 7: All variables	0.512 (0.374-0.644)
	Set 1: Saliva	0.533 (0.394-0.666)
	Set 2: Basic information	0.700 (0.573-0.824)
	Set 3: Basic information + Oral health	0.600 (0.460-0.736)
	Set 4: Basic information + Oral diseases	0.757 (0.653-0.855)
	Set 5: Basic information + Systemic diseases	0.653 (0.522-0.771)
	Set 6: Basic information + Medications	0.654 (0.529-0.768)
	Set 7: All variables	0.736 (0.615-0.842)

Set 1: Salivary flow rate and xerostomia (UFR, SFR, subjective xerostomia);

Set 2: Basic demographics + Set 1;

Set 3: Set 2 + oral health characteristics (pH, buffering capacity, VAS score, sticky saliva, oral hygiene, calculus);

Set 4: Set 2 + oral diseases (stomatitis, RAU, candidiasis, etc.);

Set 5: Set 2 + systemic diseases (hypertension, diabetes, CVD, etc.);

Set 6: Set 2 + medications (antihypertensives, psychiatric drugs, etc.);

Set 7: All variables from Sets 1–6 combined

This table reports the Area Under the Receiver Operating Characteristic Curve (AUC) and 95% confidence intervals (CIs) for four transformer-based deep learning models: FT-Transformer, TabTransformer, SAINT, and TabNet, trained on a curated feature set comprising demographic and salivary indicators (B, C, F, G, H) and key oral lesion features (O–V). Among these models, the FT-Transformer achieved the highest AUC of 0.761 (95% CI: 0.662–0.855), slightly outperforming TabTransformer (0.757) and SAINT (0.752). These three models demonstrated comparable and strong predictive capabilities, confirming the effectiveness of transformer-based architectures for tabular clinical data. In contrast, TabNet showed a markedly lower AUC of 0.596, highlighting its limited performance in this context compared to its transformer counterparts. Interestingly, the Stacking Ensemble model, which combines the outputs from XGBoost, Random Forest, TabTransformer, and FT-Transformer, yielded an AUC of 0.757, which is slightly lower than that of the FT-Transformer alone. This result suggests that ensembling may not always enhance performance, particularly when the strongest individual model (FT-Transformer) effectively captures the underlying feature relationships. This also indicates that the inclusion of models with relatively lower standalone performances (TabNet) may dilute the benefits of ensemble learning.

Table 4. DeLong test results comparing baseline and expanded feature sets: Set 2 (Basic information) vs. Set 4 (Basic information + Oral diseases)

Model Category	Model Name	P-value
ML Model	XGBoost	0.116
	CatBoost	0.361
	SVM	0.419
	Random Forest	0.703
NN Model	Tab Transformer	0.226
	FT Transformer	0.049*
	SAINT	0.106
	TabNet	0.318
Stacking Ensemble		0.489

This table presents the DeLong test-based p-values for AUC performance comparisons between Set 2 (sex, age, Xerostomia_subjective, UFR, and SFR) and the expanded features of Set 4 (O_Stomatitis, O_RAU, O_Candidiasis, O_Periodontitis, O_LichenPlanus, O_Sialodochitis, O_Glossodynia, O_BMS). The FT-Transformer model exhibited a statistically significant improvement ($p = 0.049$), highlighting its sensitivity to detailed oral health indicators. The SAINT model also approached significance ($p = 0.106$), suggesting a partial benefit. In contrast, the four conventional machine learning models did not demonstrate significant improvements, with p-values > 0.1 . These results suggest that transformer-based neural architectures are better suited for capturing the complex interactions introduced by expanded oral health features.

Table 5. DeLong test results comparing top models to TabNet, the model with the lowest predictive performance for halitosis in Set 4 (Basic information + Oral diseases)

Model		P-value
SAINT (NN)	TabNet (NN)	0.026*
FT Transformer (NN)	TabNet (NN)	0.027*
Tab Transformer (NN)	TabNet (NN)	0.055
XGBoost (ML)	TabNet (NN)	0.067
SVM (ML)	TabNet (NN)	0.198
CatBoost (ML)	TabNet (NN)	0.235
RandomForest (ML)	TabNet (NN)	0.246
CatBoost (ML)	FT Transformer (NN)	0.256
Stacking Ensemble (ML + NN)	TabNet (NN)	0.092

This table reports the p-values calculated using DeLong's test, comparing the AUC performance of TabNet with that of various conventional machine learning (ML) and neural network (NN) models with the expanded best variable set:

Sex, age, Xerostomia_subjective, UFR, SFR, O_Stomatitis, O_RAU, O_Candidiasis, O_Periodontitis, O_LichenPlanus, O_Sialodochitis, O_Glossodynia, O_BMS

The AUC was significantly different ($p = 0.027$) between the FT-Transformer model and TabNet. Similarly, SAINT achieved a significant result ($p = 0.026$), suggesting meaningful performance superiority over TabNet. Although models such as TabTransformer ($p = 0.055$) and XGBoost ($p = 0.067$) approached significance, they did not meet the conventional threshold ($p < 0.05$) for statistical significance. All other comparisons (CatBoost, Random Forest, and SVM) yielded p-values > 0.1 , indicating no statistically significant difference with respect to TabNet. These findings reinforce that transformer-based deep learning models, especially FT-Transformer and SAINT, are not only more expressive but also statistically superior in modeling complex relationships for halitosis prediction while using enriched oral health features. By contrast, TabNet exhibited a competitive but statistically weaker performance in this setting.

Declaration

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Informed consent

Informed consent was obtained from all participants involved in the study.

Author contributions

Writing and original draft preparation, Y-HL, S-WJ, and SJ; conceptualization, Y-HL, S-WJ, and SJ; methodology, Y-HL, T-SK, S-WJ, and SJ; software, Y-HL, S-WJ, and SJ; validation and formal analysis, Y-HL and SJ; investigation, Y-HL, T-SK, and SJ; resources, Y-HL, SJ, and TSK; data curation, Y-HL; writing, review, and editing, Y-HL and SJ; visualization, Y-HL and SJ; supervision, Y-HL; project administration, Y-HL; funding acquisition, Y-HL. All authors contributed to and approved the submitted manuscript.

Conflict of interest

The authors declare that this study was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

Data availability

The datasets used and/or analyzed in the current study are available from the corresponding author upon reasonable request. All experimental procedures and codes used in this study are available at the following online repository: [<https://github.com/seongwoojang1123/Neural-Network-and-Machine-Learning-Prediction-of-Halitosis-from-Oral-and-Systemic-Health-Data.git>].

Ethics statement

The procedures for human subjects in this study were conducted in accordance with the ethical standards of the Committee on Human Experimentation of our institution and the 1975 Declaration of Helsinki. This study was approved by the Ethics Review Board of Kyung Hee University Dental Hospital (IRB No. KH-DT23022). Informed consent was obtained from all participants.

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